RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Friday, 25 September 2026, 9:20 AM
State	Finished
Completed on	Friday, 25 September 2026, 9:24 AM
Time taken	3 mins 50 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>
int findDuplicate(const int* nums, int n) {
    int low = 1, high = n;

    while (low < high) {
        int mid = low + (high - low) / 2;
        int count = 0;
        for (int i = 0; i < n; i++) {
            if (nums[i] <= mid) {
                count++;
            }
        }
        if (count > mid) {
            high = mid;
        } else {
            low = mid + 1;
        }
    }

    return low;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 0;
    }
    int *nums = (int *)malloc(n * sizeof(int));
    if (nums == NULL) {
        return 1;
    }
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }

    int result = findDuplicate(nums, n);
    printf("%d\n", result);
    free(nums);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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